

YUGO ONISHI

Incoming Leinweber Postdoctoral Fellow, Stanford University
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APPOINTMENTS & RESEARCH EXPERIENCE

Sep 2026-	Incoming Leinweber Postdoctoral Fellow Stanford University
Aug-Dec 2025	KITP Graduate Fellow Kavli Institute for Theoretical Physics, UC Santa Barbara · Faculty mentor: Leon Balents
2022-2026	PhD Researcher Massachusetts Institute of Technology · Advisor: Liang Fu
Apr-Sep 2022	Doctoral Student University of Tokyo · Advisor: Takahiro Morimoto
2020-2022	Master's Researcher University of Tokyo · Advisor: Naoto Nagaosa
2019-2020	Undergraduate Researcher University of Tokyo · Advisors: Takasada Shibauchi and Kenichiro Hashimoto

EDUCATION

2026	PhD in Physics Massachusetts Institute of Technology
2022	ME in Applied Physics University of Tokyo
2020	BE in Applied Physics University of Tokyo

PUBLICATIONS

- Yugo Onishi**, Nisarga Paul, Liang Fu, "Emergent Curved Space and Gravitational Lensing in Quantum Materials." Physical Review B, 2026.
- Chung-Tao Chou, **Yugo Onishi**, Kseniia Mosina, Qiuyuan Wang, Mingda Li, Zdenek Sofer, Liang Fu, Luqiao Liu, "Giant Nonlinear Transport Response in a Magnetic Semiconductor Induced by Electrostatic Gating." Newton, 2025.
- Yugo Onishi**, Alexander Avdoshkin, Liang Fu, "Geometric Bound on the Structure Factor and a Harmonic Condition on Band Geometry." Physical Review B, 2025.
- Yugo Onishi**, Liang Fu, "Quantum Weight: A Fundamental Property of Quantum Many-Body Systems." Physical Review Research, 2025.
- Yugo Onishi**, Hiroki Isobe, Atsuo Shitade, Naoto Nagaosa, "Theory of Thermopolarization Effect." Nano Letters, 2025.
- Barun Ghosh, **Yugo Onishi**, Su-Yang Xu, Hsin Lin, Liang Fu, Arun Bansil, "Probing Quantum Geometry through Optical Conductivity and Magnetic Circular Dichroism." Science Advances, 2024.
- Yugo Onishi**, Liang Fu, "Topological Bound on the Structure Factor." Physical Review Letters, 2024.
- Yugo Onishi**, Liang Fu, "Universal Relation between Energy Gap and Dielectric Constant." Physical Review B, 2024.
- Lujin Min, Yang Zhang, Zhijian Xie, Sai Ayyagari, Leixin Miao, **Yugo Onishi**, Seng Lee, Yu Wang, Nasim Alem, Liang Fu, Zhiqiang Mao, "Colossal Room-Temperature Non-Reciprocal Hall Effect." Nature Materials, 2024.
- Anyuan Gao, Shao-Wen Chen, Barun Ghosh, Jian-Xiang Qiu, Yu-Fei Liu, **Yugo Onishi**, Chaowei Hu, Tiema Qian, Damien Berube, Thao Dinh, Houchen Li, Christian Tzschaschel, Seunghyun Park, Tianye Huang, Shang-Wei Lien, Zhe Sun, Sheng-Chin Ho, Bahadur Singh, Kenji Watanabe, Takashi Taniguchi, David Bell, Arun Bansil, Hsin Lin, Tay-Rong Chang, Amir Yacoby, Ni Ni, Liang Fu, Qiong Ma, Su-Yang Xu, "An Antiferromagnetic Diode Effect in Even-Layered MnBi₂Te₄." Nature Electronics, 2024.
- Yugo Onishi**, Liang Fu, "High-Efficiency Energy Harvesting Based on a Nonlinear Hall Rectifier." Physical Review B, 2024.
- Yugo Onishi**, Liang Fu, "Fundamental Bound on Topological Gap." Physical Review X, 2024.
- Anyuan Gao, Yu-Fei Liu, Jian-Xiang Qiu, Barun Ghosh, Thais V., **Yugo Onishi**, Chaowei Hu, Tiema Qian, Hung-Ju Tien, Shao-Wen Chen, Mengqi Huang, Damien Berube, Houchen Li, Christian Tzschaschel, Thao Dinh, Zhe Sun, Sheng-Chin Ho, Shang-Wei Lien, Bahadur Singh, Kenji Watanabe, Takashi Taniguchi, David Bell, Hsin Lin, Tay-Rong Chang, Chunhui Du, Arun Bansil, Liang Fu, Ni Ni, Peter Orth, Qiong Ma, Su-Yang Xu, "Quantum Metric Nonlinear Hall Effect in a Topological Antiferromagnetic Heterostructure." Science, 2023.
- Yugo Onishi**, Hikaru Watanabe, Takahiro Morimoto, Naoto Nagaosa, "Effects of Relaxation on the Photovoltaic Effect and Possibility for Photocurrent within the Transparent Region." Physical Review B, 2022.

15. **Yugo Onishi**, Takahiro Morimoto, Naoto Nagaosa, "Theory of Shift Heat Current and Its Application to Electron-Phonon Coupled Systems." Physical Review B, 2022.
16. Kousuke Ishida, **Yugo Onishi**, Masaya Tsujii, Kiyotaka Mukasa, Mingwei Qiu, Mikihiro Saito, Yuichi Sugimura, Kohei Matsuura, Yuta Mizukami, Kenichiro Hashimoto, Takasada Shibauchi, "Pure Nematic Quantum Critical Point Accompanied by a Superconducting Dome." Proceedings of the National Academy of Sciences, 2022.
17. K. Mukasa, K. Matsuura, M. Qiu, M. Saito, Y. Sugimura, K. Ishida, M. Otani, Y. Onishi, Y. Mizukami, K. Hashimoto, J. Gouchi, R. Kumai, Y. Uwatoko, T. Shibauchi, "High-Pressure Phase Diagrams of FeSe_{1-x}Te_x: Correlation between Suppressed Nematicity and Enhanced Superconductivity." Nature Communications, 2021.
18. **Yugo Onishi**, Hiroki Isobe, Naoto Nagaosa, "Theory of Seebeck Ratchet in a Noncentrosymmetric Electron-Phonon Coupled System." Physical Review B, 2021.

TALKS & PRESENTATIONS

May 2026	Repulsive Casimir Effect and Quantum Geometric Force Talk · Solvay Workshop "Kahler geometry: From topological quantum matter to quantum field theories" · Solvay Institute, Belgium
Mar 2026	Ground state geometry, Structure factor, and Topological bounds Talk · APS Global Physics Summit 2026 · Denver, Colorado
Mar 2026	Emergent curved space in quantum materials Talk · APS Global Physics Summit 2026 · Denver, Colorado
Jan 2026	Bound on energy gap in quantum many-body systems Seminar · University of Tokyo · University of Tokyo, Japan
Nov 2025	Bound on energy gap in quantum many-body systems Seminar · Stanford University · Stanford, California
Sep 2025	Universal bounds in quantum many-body systems Seminar · Condensed matter seminar at UC Berkeley · Berkeley, California
Sep 2025	Universal bounds in quantum many-body systems Seminar · Lunch Seminar at Caltech · Pasadena, California
Sep 2025	Universal bounds and geometry in quantum ground states Seminar · Seminar in Eslam Khalaf group at Harvard Univesity · Cambridge, Massachusetts
Apr 2025	Ground state geometry and Topological bound Seminar · Online Seminar · Online
Mar 2025	Quantum geometry and topological bound Talk · 2025 Spring Meeting, The Physical Society of Japan · Online
Mar 2025	Topological Bound on Structure Factor Talk · APS Global Physics Summit 2025 · Anaheim, California
Feb 2025	Repulsive Casimir force from Quantum Geometry Talk · The Institute for Solid State Physics, The University of Tokyo · University of Tokyo, Japan
Jan 2025	Topology and Geometry in Structure factor Seminar · ETH Zurich · ETH Zurich, Switzerland
2024	Fundamental bound on topological gap Talk · APS March Meeting 2024 · Minneapolis, Minnesota
2024	Quantum geometry, Optical property, and Energy gap Seminar · National Cheng Kung University · Tainan, Taiwan
2024	Quantum geometry, Optical property, and Energy gap Seminar · National Taiwan University · Taipei, Taiwan
2024	Quantum geometry, Optical property, and Energy gap Seminar · Academia Sinica · Taipei, Taiwan
Jul 2024	Universal bound in topological phase Talk · Quantum Designer Physics 2024 · San Sebastian, Spain
May 2024	Fundamental bound on topological gap Talk ·  workshop at Kyoto University · Kyoto University, Japan
May 2024	Fundamental gap bound in insulators Seminar · Seminar at Fujimoto group at Osaka University · Osaka University, Japan

May 2024	Fundamental gap bound in insulators Talk · NQCM2024 · University of Tokyo, Japan
Apr 2024	Fundamental bound on topological gap Seminar · ETH Zurich · ETH Zurich, Switzerland
Apr 2024	Quantum weight Seminar · Speakers' Corner Talk · Online
Apr 2024	Fundamental gap bound in insulators Seminar · Group meeting of Narang group at UCLA · Online
Mar 2024	Fundamental gap abound in insulators Seminar · Harvard CMT Kids' Seminar · Cambridge, Massachusetts
Mar 2024	Fundamental bound on topological gap Talk · APS March Meeting 2024 · Minneapolis, Minnesota
Feb 2024	Quantum geometry, Optical property, and Energy gap Seminar · National Cheng Kung University · Tainan, Taiwan
Feb 2024	Quantum geometry, Optical property, and Energy gap Seminar · National Taiwan University · Taipei, Taiwan
Feb 2024	Quantum geometry, Optical property, and Energy gap Seminar · Academia Sinica · Taipei, Taiwan
2023	High-efficiency quantum rectenna for millimeter wave and terahertz technologies Talk · APS March Meeting 2023 · Las Vegas, Nevada
Mar 2023	High-efficiency quantum rectenna for millimeter wave and terahertz technologies Talk · APS March Meeting 2023 · Las Vegas, Nevada

TEACHING

Apr 2021	Teaching Assistant for Statistical Mechanics 3 University of Tokyo, Department of Applied Physics · Tokyo, Japan
Sep 2020	Teaching Assistant for Mathematics 1D University of Tokyo, Department of Applied Physics · Tokyo, Japan
Sep 2020	Teaching Assistant for Mathematics 1D University of Tokyo, Department of Applied Physics · Tokyo, Japan
Apr 2020	Teaching Assistant for Statistical Mechanics 3 University of Tokyo, Department of Applied Physics · Tokyo, Japan

SERVICE & LEADERSHIP

2023-2025	President Japanese Association of MIT
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FELLOWSHIPS & SCHOLARSHIPS

2026-2029	Leinweber Fellowship
2025	KITP Graduate Fellowship Kavli Institute for Theoretical Physics
2022-2024	Funai Overseas Scholarship
2022	JSPS Research Fellowship for Young Scientists (DC1)
2021-2022	MERIT-WINGS Fellowship University of Tokyo